AI-Based Multimodal Retinal & Clinical Analysis System

Multimodal Retinal & Clinical Assessment Report — v1.0.0 (Research Edition)

Patient / Sample ID:	TEST_API_PATIENT	**Report ID:**	REP-TEST_API_PAT-20260829111854
Assessment Date:	2026-08-29 11:18:54	**Input Modalities:**	OCTA

1. Patient Clinical Profile & Biomarkers

Age / Group	O_CD	**Gender**	Male (1)
Education (Years)	16.0	**BMI (kg/m²)**	26.5
Hypertension (HTN)	Positive (1)	**Diabetes (DM2)**	Negative (0)
Smoking History	History (1)	**Alcohol History**	History (1)

2. Retinal Image Technical Quality (Phase 3)

Modality	Status	Quality Score	Technical Decision
OCTA	Available	85.0 / 100	**ACCEPT**
OCTB	Not available	—	—
FUNDUS	Not available	—	—

3. Multi-Task Disease Predictions & Uncertainty Analysis

Assessment Parameter	Stroke Target	Alzheimer's Disease Target
Model Prediction Class	Class 1 (Positive)	Class 1 (Positive)
Predicted Probability	**0.5418**	**0.5302**
Research Risk Category	**MODERATE RISK**	**MODERATE RISK**
Model Confidence (Phase 9)	98.94% (HIGH CONFIDENCE)	99.43% (HIGH CONFIDENCE)
Predictive Uncertainty (Variance)	$\sigma^2 = 0.0010$ (LOW)	$\sigma^2 = 0.0010$ (LOW)
Predictive Shannon Entropy	$H(p) = 0.6929$ nats	$H(p) = 0.6899$ nats

4. Retinal Visual Explainability (Swin Grad-CAM)

Grad-CAM heatmaps highlight salient retinal regions contributing to model logits. Each panel displays **[Original Retinal Scan | Grad-CAM Heatmap | Alpha-Blended Overlay]**.

Retinal Grad-CAM visualizations not available for this session.

5. Clinical Feature Attributions (Game-Theoretic SHAP)

Shapley values (SHAP) quantify marginal additive feature contributions to model log-odds.

Feature Name	Patient Value	Stroke SHAP (Impact)	Alzheimer's SHAP (Impact)

EtOH_ever	1	-0.0427 (DECREASES_RISK)	-0.0137 (DECREASES_RISK)
Gender	1	-0.0340 (DECREASES_RISK)	+0.0098 (INCREASES_RISK)
DM2	0	+0.0274 (INCREASES_RISK)	-0.0231 (DECREASES_RISK)
Old groups	O_CD	-0.0234 (DECREASES_RISK)	+0.0043 (INCREASES_RISK)
Education	16.0	+0.0185 (INCREASES_RISK)	-0.0034 (DECREASES_RISK)
BMI	26.5	-0.0166 (DECREASES_RISK)	+0.0166 (INCREASES_RISK)
Obese	0.0	+0.0137 (INCREASES_RISK)	-0.0213 (DECREASES_RISK)
HTN	1	-0.0082 (DECREASES_RISK)	-0.0092 (DECREASES_RISK)

6. Unified Multimodal Synthesis & Findings

Multimodal AI assessment was performed for patient 'TEST_API_PATIENT' utilizing available retinal modalities (OCTA) and patient tabular clinical health variables. For the Stroke target, the model estimated a predicted probability of 0.5418 (MODERATE RISK), with an estimated model confidence of 98.9% and low predictive uncertainty (variance=0.0010). For the Alzheimer's Disease target, the model estimated a predicted probability of 0.5302 (MODERATE RISK), with an estimated model confidence of 99.4% and low predictive uncertainty (variance=0.0010). Model explainability heatmaps (Grad-CAM) and game-theoretic clinical feature contributions (SHAP) are integrated to highlight computational saliency patterns.

7. Research Limitations & Methodological Constraints

1. Experimental Model Scope: Algorithms are trained on curated academic datasets and require prospective clinical validation.
2. Biological Complexity: Retinal microvascular and neurodegenerative patterns reflect multi-factorial systemic processes.
3. Explainability Attribution: Grad-CAM heatmaps and SHAP values indicate model parameter attributions and do not establish medical causality.
4. Modality Missingness: Missing imaging modalities degrade fusion representational capacity relative to complete tripartite scans.

MANDATORY RESEARCH DISCLAIMER:

IMPORTANT RESEARCH NOTICE: This report is generated by an experimental multimodal artificial intelligence research prototype for investigational and decision-support purposes only. The predictions, probability estimates, uncertainty metrics, and explainability visualizations do not constitute a clinically verified medical diagnosis and must never substitute for professional evaluation, diagnosis, or treatment by a board-certified healthcare provider.